**DSA06 DATA HANDLING AND VISUALIZATION | Activity: Correlation Detective | Slot C**

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*Time: 20 to 25 minutes Total: 20 marks Tool: R (ggplot2 / base R). Work alone; your case file is different from your neighbour's. Write your answers and paste your plots and code.*

## Case File CD-C06: The Hospital ward Case [20 marks]

**Briefing:** You are the data detective hired by the nursing superintendent of a district hospital. The file holds 14 daily records from the hot weeks, with seven variables. Some relationships can be trusted. Others are traps: a hidden driver, a curve that fools r, and a single wrong record. Find them.

| R name | What it measures | R name | What it measures |
|---|---|---|---|
| outside_temp | Outside temperature (deg C) | discomfort | Patient discomfort score |
| cold_drinks | Canteen cold-drink sales | nurses | Nurses on shift |
| heat_admits | Heat-related admissions | med_errors | Medication errors reported |
| ward_dev | Ward temperature minus 21 deg C | | |

**Evidence: run this code in R to create your case file (do not edit it).**

```
set.seed(1247649)
n <- 14
outside_temp <- round(rnorm(n, 35.5, 2.9), 1)
cold_drinks <- round(209 + 48.0 * (0.95 * (outside_temp - 35.5) / 2.9 + 0.30 * rnorm(n)), 0)
heat_admits <- round(9 + 2.9 * (-0.88 * (outside_temp - 35.5) / 2.9 + 0.48 * rnorm(n)), 0)
ward_dev <- round(seq(-3.6, 3.6, length.out = n)[order(runif(n))], 1)
discomfort <- round(3.1 + 8.5 * (ward_dev / 3.6)^2 + rnorm(n, 0, 0.3), 1)
nurses <- round(rnorm(n, 15, 2.9), 0)
med_errors <- round(4 + 1.4 * (0.18 * (nurses - 15) / 2.9 + 0.98 * rnorm(n)), 0)
nurses[9] <- 24; med_errors[9] <- 9   # row 9 was typed in by a trainee
case <- data.frame(outside_temp, cold_drinks, heat_admits, ward_dev, discomfort, nurses, med_errors)
```

**Tasks:**

1. The matrix. Use cor(case) to get the 7 x 7 correlation matrix (round to 2 decimals). State n and how many distinct pairs of variables there are. Draw a correlogram of the matrix (ggplot2 geom_tile with geom_text, or corrplot). [4]
2. The strongest clues. Name the strongest positive pair and the strongest negative pair (off-diagonal). Give r for each and call it weak (below 0.3), moderate (0.3 to 0.7) or strong (above 0.7). [3]
3. The deceptive zero. Plot discomfort against ward_dev with a straight line, then add a quadratic fit (formula = y ~ x + I(x^2)). Report r and the R-squared of the quadratic fit. In one sentence, say why r misses this relationship. [4]
4. The outlier alibi. Plot med_errors against nurses. Find the row that looks wrong and give its row number. Report r for all rows and r without that row, and say what the single record did to r. [4]
5. The prime suspect. cold_drinks and heat_admits are strongly linked. Report r(cold_drinks, heat_admits), then r(outside_temp, cold_drinks) and r(outside_temp, heat_admits). Compute the partial correlation of cold_drinks and heat_admits controlling for outside_temp: correlate resid(lm(cold_drinks ~ outside_temp)) with resid(lm(heat_admits ~ outside_temp)). Write a two-sentence verdict: does cold_drinks cause heat_admits? Name one study design that could test causation. [5]

***Detective's rule:*** *correlation measures a straight-line link only, it can be inflated by one point, and it never proves cause.*

*Code of Conduct: I certify that this work is my own. Signature: _____*

**LAB ACTIVITY RECORD**

**Activity: Correlation Detective (Case File CD-C06)**

| | |
|---|---|
| **Department** | AIML |
| **Course Code / Course Name** | DSA0601 / Data Handling and Visualization |
| **Experiment** | Activity - Correlation Detective (R) |
| **Name** | BANDLAPALLI BHANUTEJA REDDY |
| **Register Number** | 192325016 |
| **Date** | 22 OCTOBER 2026 |
| **Day** | Thursday |

**Aim**

To investigate a hospital ward case file of 14 daily records and 7 variables using correlation: to build and draw the correlation matrix, to name the strongest relationships, and to catch the three traps hidden in the data (a curve that fools r, a single wrong record, and a hidden driver that makes two variables look linked).

**Key Ideas**

| Term | Meaning in simple words |
|---|---|
| Correlation r | A number from -1 to +1 that measures how well two variables follow a straight line. Weak: below 0.3. Moderate: 0.3 to 0.7. Strong: above 0.7. |
| r-squared | r x r: the share of the ups and downs of one variable that a straight line on the other can explain. |
| Curve trap | If the link is a curve (for example a U shape), the straight-line r can be close to 0 even though the link is very strong. |
| Outlier trap | One extreme or wrongly typed record can create or destroy a correlation, because r uses every point equally. |
| Confounder (hidden driver) | A third variable that moves two others together, so they look related although neither causes the other. |
| Partial correlation | The correlation of two variables after the effect of a third variable has been removed from both. |

**The Case File**

| Variable | What it measures | Variable | What it measures |
|---|---|---|---|
| outside_temp | Outside temperature (deg C) | discomfort | Patient discomfort score |
| cold_drinks | Canteen cold-drink sales | nurses | Nurses on shift |
| heat_admits | Heat-related admissions | med_errors | Medication errors reported |
| ward_dev | Ward temperature minus 21 deg C | | |

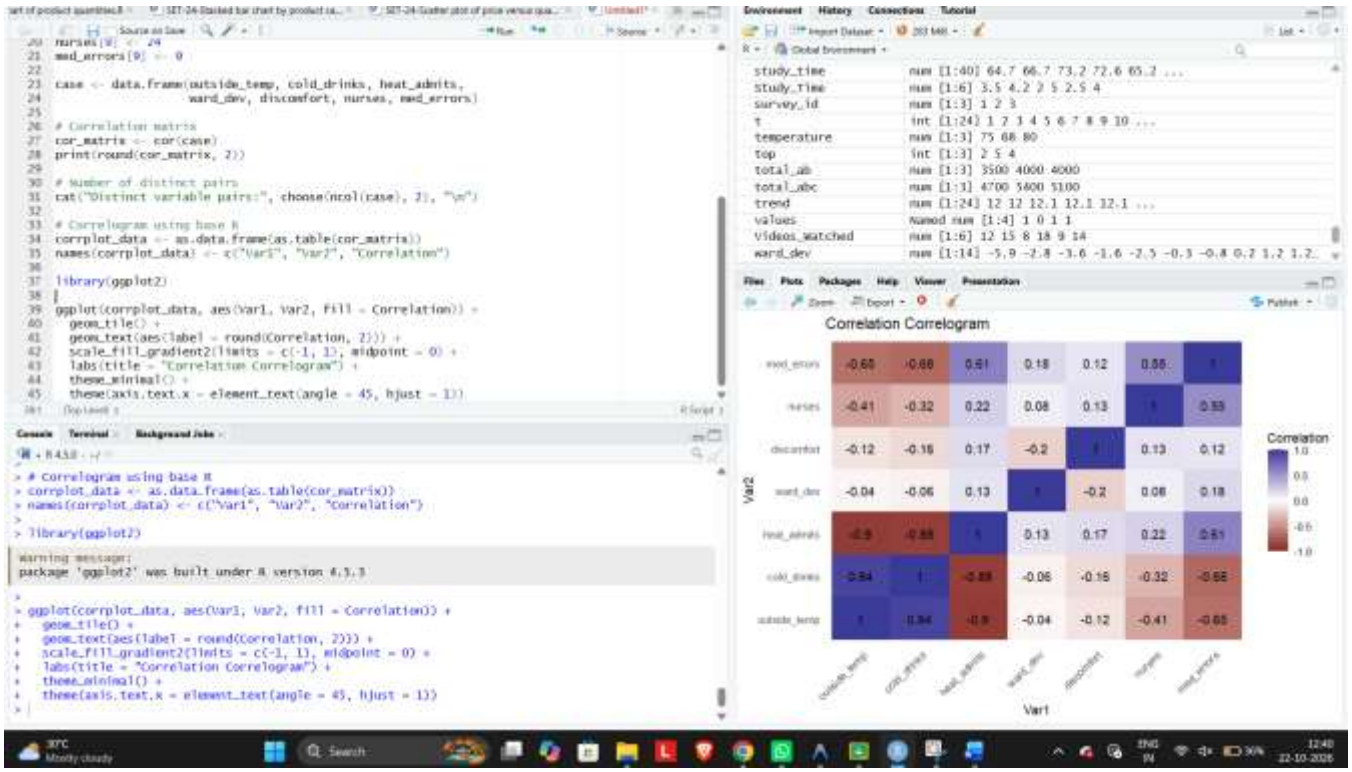
The evidence code of the question was run in R without any change (set.seed(1247649), n = 14), and every number below comes from that real output.

**Task 1: The Correlation Matrix**

**[4 marks]**

**Aim:** To compute the 7 x 7 correlation matrix with cor(case), state n and the number of distinct pairs, and draw a correlogram.

**Output**




**Working**

**Step 1:** n = 14 (14 daily records, one row for each day).

**Step 2:** Distinct pairs of variables = 7 x 6 / 2 = 21. (The matrix has 49 cells, but 7 are the diagonal, where every variable has r = 1 with itself, and the 42 others are in mirror pairs, so there are 42 / 2 = 21 different pairs.)

**Step 3:** cor(case) gives the matrix below, rounded to 2 decimals with round(cor(case), 2).

**Step 4:** The correlogram is a grid of coloured tiles with the r value written on each tile (geom_tile with geom_text). Red tiles are positive, blue tiles are negative, and white is near 0.

| | temp | cold | heat | ward | disc | nurs | errs |
|---|---|---|---|---|---|---|---|
| outside_temp | 1.00 | 0.94 | -0.90 | -0.38 | 0.24 | -0.01 | -0.18 |
| cold_drinks | 0.94 | 1.00 | -0.88 | -0.34 | 0.29 | -0.09 | -0.13 |
| heat_admits | -0.90 | -0.88 | 1.00 | 0.27 | -0.26 | -0.02 | 0.00 |
| ward_dev | -0.38 | -0.34 | 0.27 | 1.00 | -0.05 | -0.49 | -0.04 |
| discomfort | 0.24 | 0.29 | -0.26 | -0.05 | 1.00 | -0.15 | -0.24 |
| nurses | -0.01 | -0.09 | -0.02 | -0.49 | -0.15 | 1.00 | 0.65 |
| med_errors | -0.18 | -0.13 | 0.00 | -0.04 | -0.24 | 0.65 | 1.00 |

Column heads are short for the same names as the rows: temp = outside_temp, cold = cold_drinks, heat = heat_admits, ward = ward_dev, disc = discomfort, nurs = nurses, errs = med_errors.

**Result:** n = 14 and there are 21 distinct pairs. The correlogram shows three strong tiles in the top left corner (outside_temp, cold_drinks and heat_admits) and one moderate tile for nurses and med_errors (0.65); all other tiles are weak.

**Task 2: The Strongest Clues**

**Aim:** To name the strongest positive pair and the strongest negative pair (off the diagonal) and classify each as weak, moderate or strong.

**Output**



```
positive <- which(cm == max(cm, na.rm = TRUE),
                  arr.ind = TRUE)[1, ]
cat("Strongest positive pair:\n")
print(c(rownames(cm)[positive[1]],
        colnames(cm)[positive[2]]))
print(cm[positive[1], positive[2]])

# Strongest negative correlation
negative <- which(cm == min(cm, na.rm = TRUE),
                  arr.ind = TRUE)[1, ]
cat("Strongest negative pair:\n")
print(c(rownames(cm)[negative[1]],
        colnames(cm)[negative[2]]))
print(cm[negative[1], negative[2]])

# Classify correlation strength
classify <- function(r) {
  if (abs(r) < 0.3) "Weak"
  else if (abs(r) < 0.7) "Moderate"
  else "Strong"
}

cat("Positive strength:",
    classify(cm[positive[1], positive[2]]), "\n")
cat("Negative strength:",
    classify(cm[negative[1], negative[2]]), "\n")
```

```
[1] "heat_admits"  "outside_temp"
> print(cm[negative[1], negative[2]])
[1] -0.8980434
>
> # Classify correlation strength
> classify <- function(r) {
+   if (abs(r) < 0.3) "Weak"
+   else if (abs(r) < 0.7) "Moderate"
+   else "Strong"
+ }
>
> cat("Positive strength:",
+     classify(cm[positive[1], positive[2]]), "\n")
Positive strength: Strong
> cat("Negative strength:",
+     classify(cm[negative[1], negative[2]]), "\n")
Negative strength: Strong
>
```

**Working**

**Step 1:** Ignore the diagonal (always 1.00) and the mirror cells, and look for the largest positive and the most negative value among the 21 pairs.

**Step 2:** Strongest positive pair: outside_temp and cold_drinks, r = +0.94.

**Step 3:** Strongest negative pair: outside_temp and heat_admits, r = -0.90. (The next one is cold_drinks and heat_admits, r = -0.88.)

**Step 4:** Classify with the scale of the question: |0.94| and |-0.90| are both above 0.7, so both are strong.

| Pair | r | Strength | Meaning |
|---|---|---|---|
| outside_temp and cold_drinks | +0.94 | Strong | Hotter days, more cold drinks sold |
| outside_temp and heat_admits | -0.90 | Strong | Hotter days, fewer heat admissions (surprising; see task 5) |

Other pairs worth noting: nurses and med_errors (+0.65, moderate) and ward_dev and nurses (-0.49, moderate). With only n = 14 days, an r must be larger than about 0.53 in size before it is more than chance at the 5% level, so -0.49 is not convincing, while 0.65 passes that test but is checked in task 4.

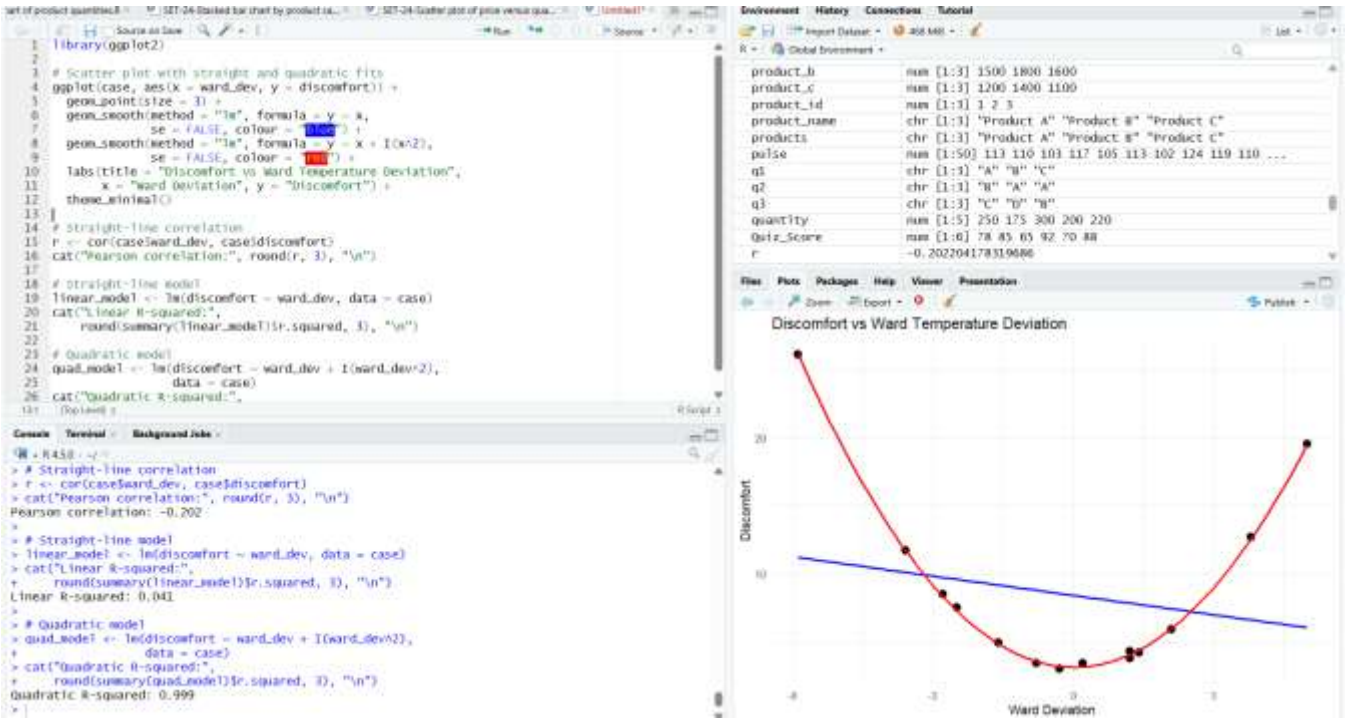
**Result:** The strongest positive pair is outside_temp and cold_drinks (r = 0.94, strong). The strongest negative pair is outside_temp and heat_admits (r = -0.90, strong).

**Task 3: The Deceptive Zero**

**[4 marks]**

**Aim:** To plot discomfort against ward_dev with a straight line and a quadratic fit, report r and the R-squared of the quadratic, and explain why r misses the relationship.

**Output**




**Working**

**Step 1:** Straight-line correlation: r(ward_dev, discomfort) = -0.05. So r-squared = 0.05 x 0.05 = 0.0025, that is only 0.25% of the variation, practically zero.

**Step 2:** Quadratic fit: lm(discomfort ~ ward_dev + I(ward_dev^2)). In an R formula the square must be written as I(x^2); without I() the ^ is read as a formula symbol.

**Step 3:** The fitted curve is discomfort = 3.26 - 0.063 x ward_dev + 0.641 x ward_dev^2. The coefficient of the squared term is highly significant (t = 63.2, p < 0.001).

**Step 4:** R-squared of the quadratic fit = 0.997, which means 99.7% of the variation of discomfort is explained by the ward temperature through a curve.

**Step 5:** The curve has its lowest point at ward_dev = 0.063 / (2 x 0.641) = 0.05, so the ward is most comfortable at about 21 deg C (ward_dev = 0), and discomfort climbs steeply when the ward is either colder or hotter than that (for example 11.6 at -3.6 and 11.4 at +3.6).

| Fit | Measure | Value |
|---|---|---|
| Straight line | r | -0.05 (r-squared = 0.0025) |
| Quadratic | R-squared | 0.997 |

**In one sentence:** r only measures straight-line links, and in this U-shaped relationship the rising right arm and the falling left arm cancel each other out, so r is near 0 although the link is almost perfect.

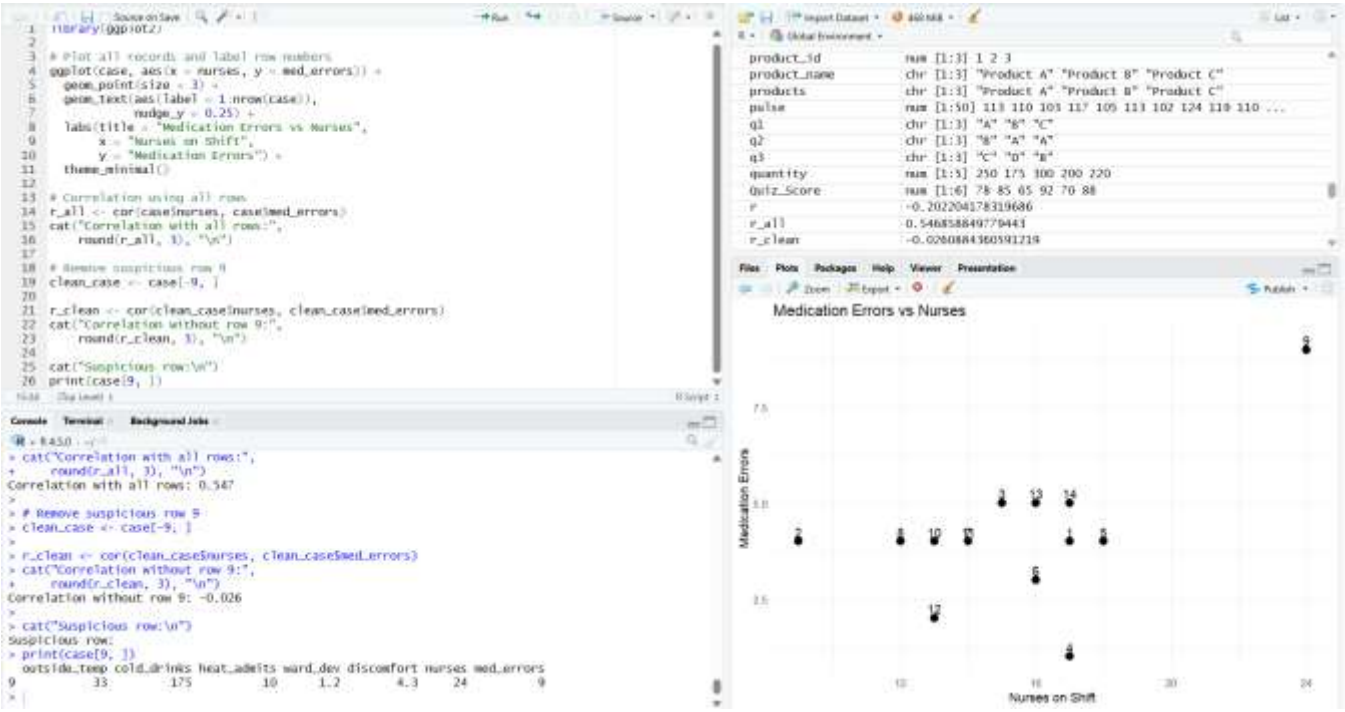
**Result:** r = -0.05, but the quadratic R-squared = 0.997. A near-zero r does not mean that there is no relationship; always plot the data.

**Task 4: The Outlier Alibi**

**[4 marks]**

**Aim:** To plot med_errors against nurses, find the row that looks wrong, and report r for all rows and without that row.

**Output**




**Working**

**Step 1:** In the plot of med_errors against nurses, 13 of the 14 points are bunched together (nurses between 12 and 18, errors between 1 and 5), and one point is far away at the top right: 24 nurses and 9 errors.

**Step 2:** That record is row 9: outside_temp 33.0, nurses = 24, med_errors = 9. The code even says "row 9 was typed in by a trainee" (nurses[9] <- 24; med_errors[9] <- 9).

**Step 3:** r for all 14 rows: r(nurses, med_errors) = +0.65, a moderate positive link.

**Step 4:** r without row 9 (the other 13 rows): r = +0.09, which is a very weak link (r-squared less than 1%).

**Step 5:** The size of the change: 0.65 - 0.09 = 0.56, so r falls by 0.56 when one record out of 14 is removed.

| Rows used | r(nurses, med_errors) | Strength |
|---|---|---|
| All 14 rows | +0.65 | Moderate |
| Without row 9 (13 rows) | +0.09 | Weak |

**What the single record did to r:** row 9 is a lone point far from the others, in the corner where the two variables are both high, so it pulls the line up to meet it and by itself creates the moderate correlation. The apparent link "more nurses, more errors" exists only because of one doubtful record.

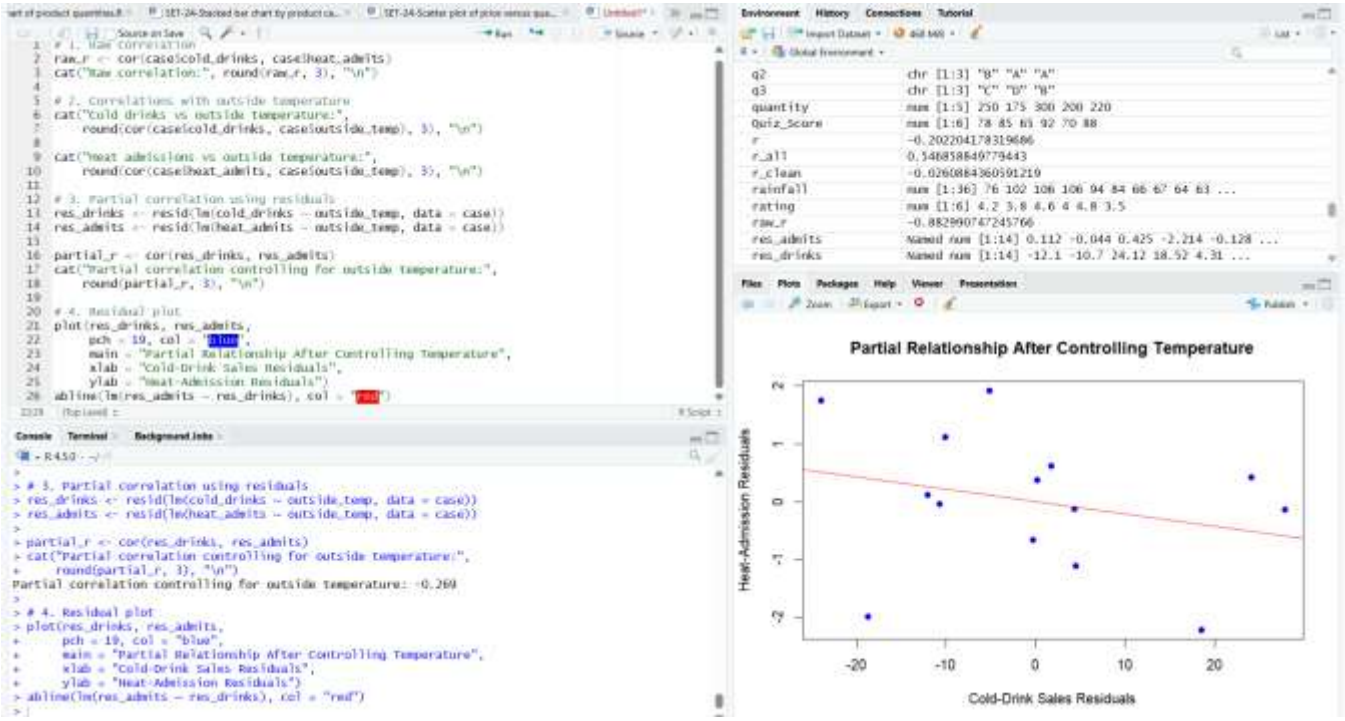
**Result:** Row 9 is the odd one out. r = 0.65 with all rows and 0.09 without row 9, so one point inflated r from weak to moderate.

**Task 5: The Prime Suspect**

**[5 marks]**

**Aim:** To compute the correlations among outside_temp, cold_drinks and heat_admits, find the partial correlation of cold_drinks and heat_admits controlling for outside_temp, and give a verdict.

**Output**




**Working**

**Step 1:** The three correlations: r(cold_drinks, heat_admits) = -0.88; r(outside_temp, cold_drinks) = +0.94; r(outside_temp, heat_admits) = -0.90.

**Step 2:** Partial correlation: take the residuals of the two regressions res_c <- resid(lm(cold_drinks ~ outside_temp)) and res_h <- resid(lm(heat_admits ~ outside_temp)). These are cold-drink sales and heat admissions with the effect of temperature removed. Then cor(res_c, res_h) = -0.27.

**Step 3:** Check with the formula: r_xy.z = (r_xy - r_xz x r_yz) / square root of [(1 - r_xz^2)(1 - r_yz^2)] = (-0.883 - 0.937 x (-0.898)) / square root of [(1 - 0.879)(1 - 0.806)] = (-0.883 + 0.842) / square root of (0.121 x 0.194) = -0.041 / 0.153 = -0.27, which agrees.

**Step 4:** So the link between cold drinks and heat admissions drops from -0.88 (r-squared 78%) to -0.27 (r-squared 7%) once temperature is held constant. With n = 14 even this small remainder is not statistically significant (t = -0.93, p about 0.37).

| Correlation | Value |
|---|---|
| r(cold_drinks, heat_admits) | -0.88 |
| r(outside_temp, cold_drinks) | +0.94 |
| r(outside_temp, heat_admits) | -0.90 |
| Partial r(cold_drinks, heat_admits \| outside_temp) | -0.27 |

**Verdict (two sentences):** No, cold_drinks does not cause heat_admits: outside temperature is the hidden driver that moves both, because hot days increase the sales of cold drinks and change the admissions, and once temperature is removed the link almost disappears (-0.88 falls to -0.27). A strong correlation only shows that two things move together, so it cannot prove that one causes the other.

**Study design that could test causation:** a randomised experiment. For example, on days of similar temperature, randomly choose which hospital wards or days will have free cold drinks offered to patients and visitors and which will not, and compare the number of heat-related admissions between the two randomised groups. Random assignment makes the temperature the same, on average, in both groups, so any difference could be attributed to the cold drinks.

**Result:** Partial r = -0.27 against the raw r = -0.88, so temperature explains the link. Cold drinks do not cause heat admissions; a randomised trial would be needed to test causation.

**Summary of Answers**

| Task | Answer |
|---|---|
| 1 | n = 14; 21 distinct pairs; matrix and correlogram drawn |
| 2 | Strongest positive: outside_temp and cold_drinks (+0.94, strong). Strongest negative: outside_temp and heat_admits (-0.90, strong) |
| 3 | Straight-line r = -0.05 but quadratic R-squared = 0.997: a U-shaped curve fools r |
| 4 | Row 9 (24 nurses, 9 errors) is the wrong record; r = 0.65 with it and 0.09 without it |
| 5 | r(cold, heat) = -0.88, r(temp, cold) = +0.94, r(temp, heat) = -0.90, partial r = -0.27: temperature is the hidden driver; test causation with a randomised experiment |

**Conclusion: Detective's Rule**

The case file contained three traps. A confounder (outside temperature) made cold drinks look as if they caused heat admissions, a curve (ward temperature and discomfort) gave r near zero although the link is almost perfect, and one wrongly typed record (row 9) created a moderate correlation between nurses and medication errors. Correlation measures a straight-line link only, it can be inflated by one point, and it never proves cause; so always plot the data, check for outliers and think about hidden drivers before drawing a conclusion.